

**Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.**

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = r'sales_data.csv'
df = pd.read_csv(file_path)

print("First few rows:")
print(df.head())

print("\nMissing values:")
print(df.isnull().sum())

df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
df = df.dropna(subset=['Product', 'Quantity', 'Region'])

print("\nSummary Statistics:")
print(df.describe())

product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print("\nProduct Summary:")
print(product_summary)

plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.xticks(rotation=45)
```

```
plt.show()

df['Date'] = pd.to_datetime(df['Date'])

sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()

plt.figure(figsize=(10, 6))

plt.plot(sales_over_time['Date'], sales_over_time['Sales'])

plt.xlabel('Date')

plt.ylabel('Total Sales')

plt.title('Sales Over Time')

plt.xticks(rotation=45)

plt.show()

pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

print("\nPivot Table:")

print(pivot_table)

correlation_matrix = df.select_dtypes(include='number').corr()

print("\nCorrelation Matrix:")

print(correlation_matrix)

plt.figure(figsize=(8, 6))

sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')

plt.title('Correlation Matrix')

plt.show()
```

## Output:

First few rows:

|   | Date       | Product   | Sales | Quantity | Region |
|---|------------|-----------|-------|----------|--------|
| 0 | 2023-01-01 | Product A | 200   | 4        | North  |
| 1 | 2023-01-02 | Product B | 150   | 3        | South  |
| 2 | 2023-01-03 | Product A | 220   | 5        | North  |
| 3 | 2023-01-04 | Product C | 300   | 6        | East   |
| 4 | 2023-01-05 | Product B | 180   | 4        | West   |

Missing values:

Date 0

Product 0

Sales 0

Quantity 0

Region 0

dtype: int64

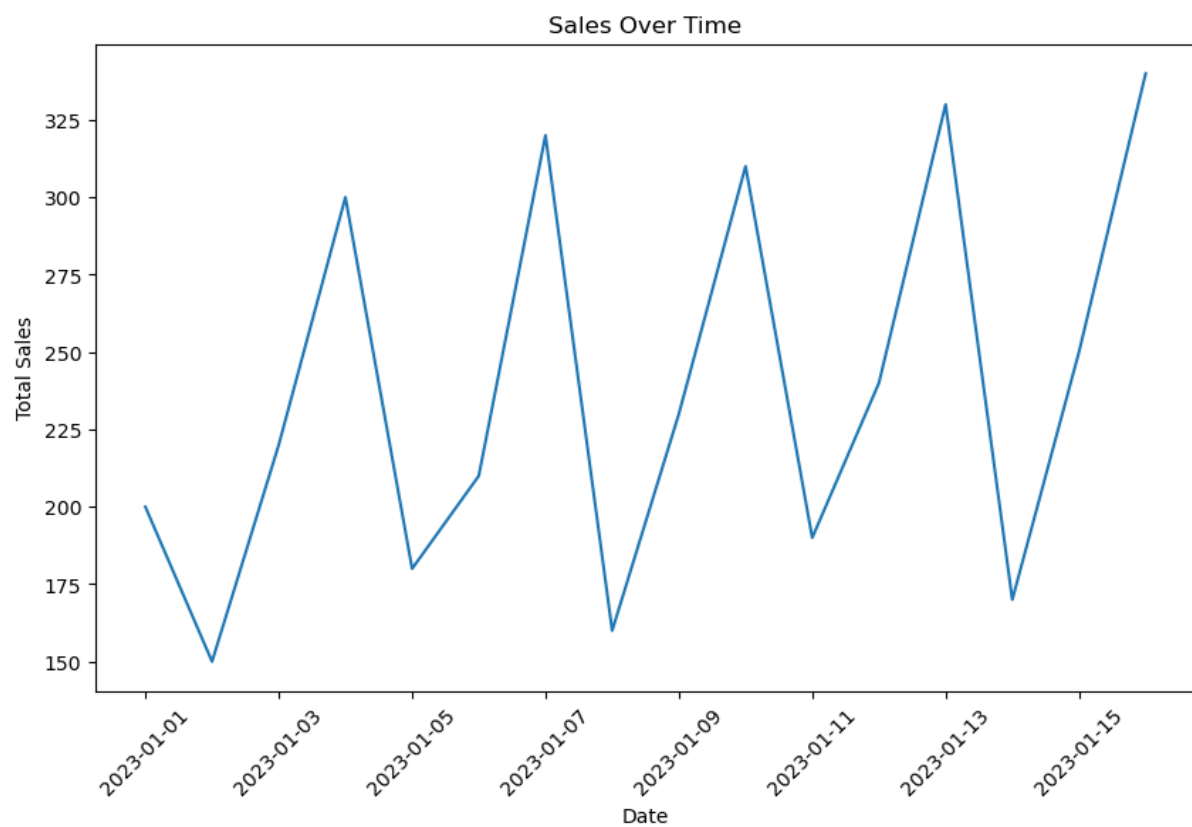
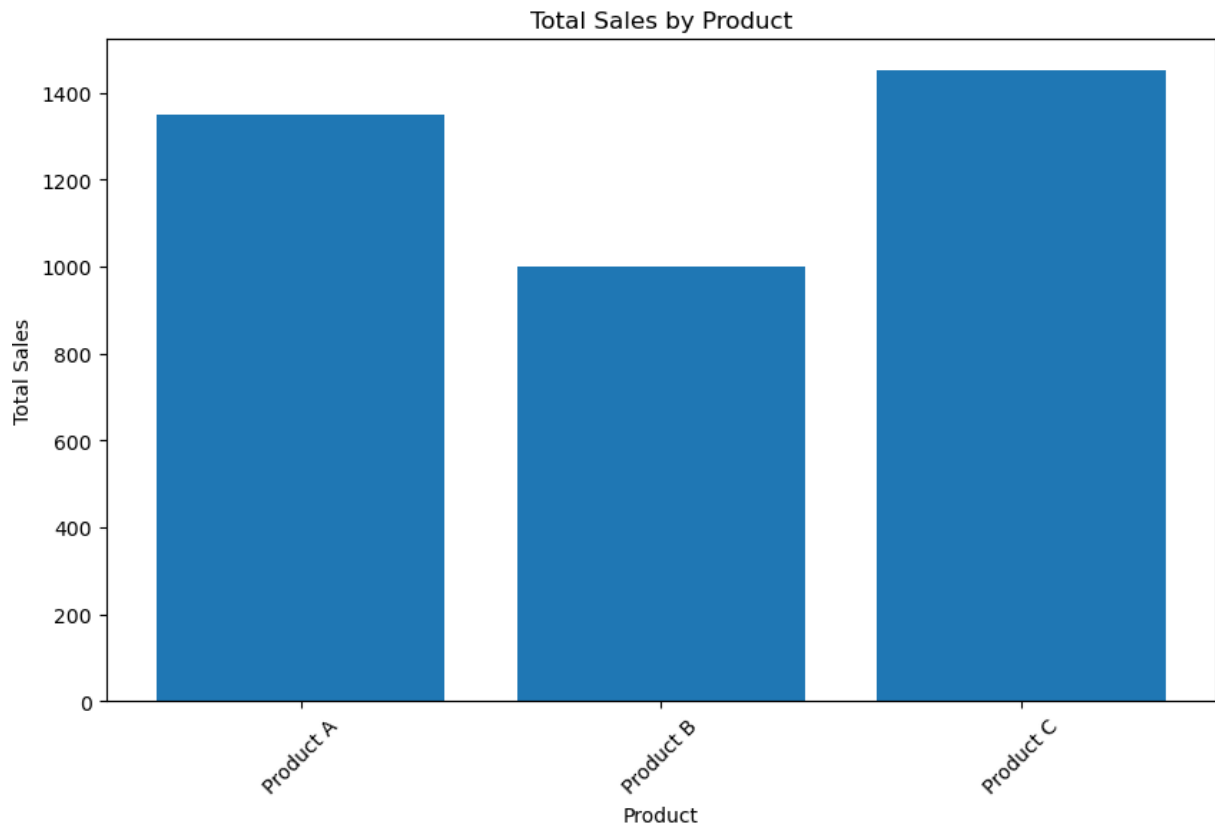
Summary Statistics:

|       | Sales      | Quantity  |
|-------|------------|-----------|
| count | 16.000000  | 16.000000 |
| mean  | 237.500000 | 5.375000  |
| std   | 64.031242  | 1.821172  |
| min   | 150.000000 | 3.000000  |
| 25%   | 187.500000 | 4.000000  |
| 50%   | 225.000000 | 5.000000  |
| 75%   | 302.500000 | 7.000000  |
| max   | 340.000000 | 8.000000  |

Product Summary:

|   | Product   | Sales | Quantity |
|---|-----------|-------|----------|
| 0 | Product A | 1350  | 31       |
| 1 | Product B | 1000  | 23       |

2 Product C 1450 32



Pivot Table:

| Product | Product A | Product B | Product C |
|---------|-----------|-----------|-----------|
|---------|-----------|-----------|-----------|

| Region |
|--------|
|--------|

|      |   |   |      |
|------|---|---|------|
| East | 0 | 0 | 1450 |
|------|---|---|------|

|       |      |   |   |
|-------|------|---|---|
| North | 1350 | 0 | 0 |
|-------|------|---|---|

|       |   |     |   |
|-------|---|-----|---|
| South | 0 | 480 | 0 |
|-------|---|-----|---|

|      |   |     |   |
|------|---|-----|---|
| West | 0 | 520 | 0 |
|------|---|-----|---|

Correlation Matrix:

|  | Sales | Quantity |
|--|-------|----------|
|--|-------|----------|

|       |          |          |
|-------|----------|----------|
| Sales | 1.000000 | 0.957592 |
|-------|----------|----------|

|          |          |          |
|----------|----------|----------|
| Quantity | 0.957592 | 1.000000 |
|----------|----------|----------|

